INCH POUND

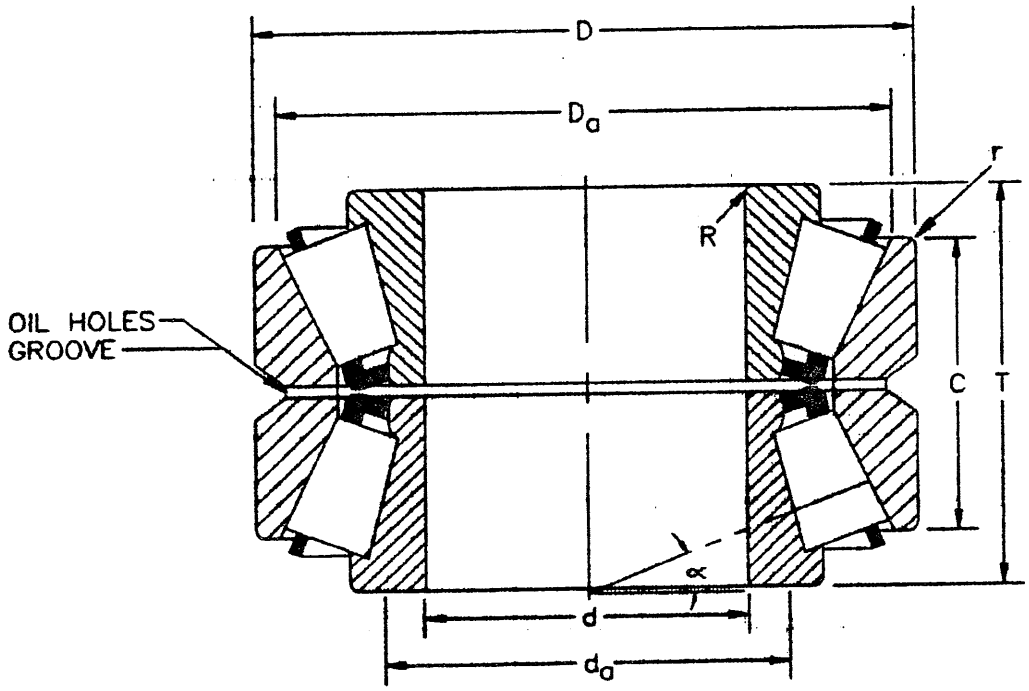
FF-B-187/7
10 SEPTEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, ROLLER, TAPERED, DOUBLE ROW, STEEP ANGLE, TWO SINGLE CONES. ONE DOUBLE CUP. (TYPE TDOS)

This specification is approved by the Administrator, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring the bearings described herein, shall consist of this document and the latest issue of FF-B-187/GEN.



BEARING CONFIGURATION

FSC 3110

AMSC N/A

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	C BEARING WIDTH OVER CUP	T CONES WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	da RECOMMENDED SHOULDER DIAMETER SHAFT	Da RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
01	21063-21226D	0.6250	2.2500	1.4375	1.9375	0.03	0.03	1.14	2.01	.99	12800
02 *	21075-21226D	0.7500	2.2500	1.4375	1.9375	0.06	0.03	1.24	2.01	.99	12800
03	43112-43319D	1.1250	3.1875	1.5625	2.1875	0.03	0.06	1.67	2.91	.87	21500
04	43125-43319D	1.2500	3.1875	1.5625	2.1875	0.06	0.06	1.73	2.91	.87	21500
05	43131-43319D	1.3125	3.1875	1.5625	2.1875	0.14	0.06	2.01	2.91	.87	21500
06	43132-43319D	1.3125	3.1875	1.5625	2.1875	0.08	0.06	1.89	2.91	.87	21500
07	44131-44363D	1.3125	3.6250	1.5625	2.1875	0.08	0.06	2.01	3.35	.75	23300
08	44143-44363D	1.4375	3.6250	1.5625	2.1875	0.09	0.06	2.13	3.35	.75	23300
09	44150-44363D	1.5000	3.6250	1.5625	2.1875	0.09	0.06	2.28	3.35	.75	23300
10	53150-53376D	1.5000	3.7500	1.7500	2.5626	0.06	0.03	2.17	3.50	.79	27700
11	44158-44363D	1.5625	3.6250	1.5625	2.1875	0.14	0.06	2.28	3.35	.75	23300
12	44162-44363D	1.6250	3.6250	1.5625	2.1875	0.09	0.06	2.24	3.35	.75	23300
13	53162-53376D	1.6250	3.7500	1.7500	2.5626	0.06	0.03	2.24	3.50	.79	27700
14	53176-53376D	1.7500	3.7500	1.7500	2.5626	0.05	0.03	2.32	3.50	.79	27700
15	53177-53376D	1.7500	3.7500	1.7500	2.5626	0.14	0.03	2.48	3.50	.79	27700
16	53178-53376D	1.7500	3.7500	1.7500	2.5626	0.08	0.03	2.36	3.50	.79	27700
17 *	53176-53390D	1.7500	3.8750	1.7500	2.5626	0.05	0.06	2.32	3.56	.79	27700
18	55175-55433D	1.7500	4.3300	1.6876	2.5000	0.14	0.02	2.64	4.13	.66	29500
19	55187-55433D	1.8750	4.3300	1.6876	2.5000	0.14	0.02	2.72	4.13	.66	29500
20	72188C-72488D	1.8750	4.8750	2.1875	3.0625	0.03	0.06	2.72	4.53	.79	50000

* INACTIVE FOR NEW DESIGN AFTER 13 JAN 1982.

1/ PART NUMBERS ARE FOR REFERENCE ONLY. PART NUMBERS REFLECT THE CUP AND CONE NUMBERS USED BY INDUSTRY AND THE AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA).

2/ THESE MAXIMUM FILLET RADII SHALL BE CLEARED BY THE BEARING CORNERS.

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TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	C BEARING WIDTH OVER CUP	T CONES WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	da RECOMMENDED SHOULDER DIAMETER SHAFT	Da RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
21	55200-55433D	2.0000	4.3300	1.6876	2.5000	0.14	0.02	2.80	4.13	.66	29500
22	55200-55444D	2.0000	4.4375	1.8125	2.5625	0.14	0.06	2.80	4.13	.66	29400
23	66200-66462D	2.0000	4.6250	2.1250	2.8750	0.14	0.03	2.80	4.37	.93	41200
24	72200-72488D	2.0000	4.8750	2.1875	3.0625	0.14	0.06	2.91	4.53	.79	50000
25	72200C-72488D	2.0000	4.8750	2.1875	3.0625	0.14	0.06	3.03	4.53	.79	50000
26	72201C-72488D	2.0000	4.8750	2.1875	3.0625	0.03	0.06	3.03	4.53	.79	50000
27	55206-55444D	2.0625	4.4375	1.8125	2.5625	0.14	0.06	2.83	4.13	.66	29400
28	66212-66462D	2.1250	4.6250	2.1250	2.8750	0.14	0.03	2.87	4.37	.93	41200
29	72212C-72488D	2.1250	4.8750	2.1875	3.0625	0.14	0.06	3.11	4.53	.79	50000
30	66584-66522D	2.1250	5.1174	1.8750	2.7500	0.14	0.03	2.95	4.65	.88	42800
31	72218C-72488D	2.1875	4.8750	2.1875	3.0625	0.14	0.06	3.15	4.53	.79	50000
32	66225-66462D	2.2500	4.6250	2.1250	2.8750	0.14	0.03	2.99	4.37	.93	41200
33	72225C-72488D	2.2500	4.8750	2.1875	3.0625	0.14	0.06	3.19	4.53	.79	50000
34	78225-78549D	2.2500	5.5000	2.0395	3.0625	0.14	0.06	3.27	5.16	.67	55200
35	66585-66522D	2.3622	5.1174	1.8750	2.7500	0.14	0.03	3.11	4.65	.88	42800
36	78250-78549D	2.5000	5.5000	2.0395	3.0625	0.09	0.06	3.35	5.16	.67	55200
37	9285-92200	3.0000	6.3750	2.7810	4.1560	0.14	0.03	4.06	6.03	.82	81700
38 •	9378-93200	3.0000	7.0000	2.9375	4.3125	0.14	0.09	4.13	6.46	.76	84200
39	9380-9320D	3.0000	7.0000	2.9375	4.3125	0.14	0.09	4.13	6.46	.76	49100
40	9385-9320D	3.3125	7.0000	2.9375	4.3125	0.14	0.09	4.37	6.46	.76	84200

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TABLE I. DASH NUMBERS AND DIMENSIONS – CONTINUED

DASH NUMBER	PART NUMBER 1/ CONE–CUP	d BORE	D OUTSIDE DIA	C BEARING WIDTH OVER CUP	T CONES WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	da RECOMMENDED SHOULDER DIAMETER SHAFT	Da RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
41	9386H–9320D	3.3125	7.0000	2.9375	4.3125	0.14	0.09	4.37	6.46	.76	84200
42 *	98350–98789D	3.5000	7.8750	3.1581	4.5625	0.14	0.09	4.65	7.40	.92	111700
43	HM921343–HM9213100	3.5400	7.4790	2.1300	3.3800	0.14	0.06	4.61	7.13	.67	79600
44	M919045–M919010D	3.5425	6.3740	1.7500	2.7500	0.14	0.06	4.29	6.06	.80	56700
45	HM821547–HM821511D	3.9362	7.7500	2.9300	4.0700	0.14	0.06	4.84	7.36	.96	103700
46	98400–98789D	4.0000	7.8750	3.1581	4.5625	0.14	0.09	5.04	7.40	.92	111700
47	37425–37626D	4.2500	6.2500	1.5625	2.1251	0.14	0.03	4.80	5.98	.96	32400
48	37431–37626D	4.3125	6.2500	1.5625	2.1251	0.14	0.03	4.84	5.98	.96	32400
49	H924045–H924010D	4.3750	8.4375	3.3125	4.5625	0.14	0.06	5.47	8.07	.87	129100
50	HH924349–HH924310D	4.3750	9.5000	4.2500	6.2500	0.25	0.06	6.18	8.90	.80	184000
51 *	HM926740–HM9267100	4.5000	9.0000	3.3125	4.5625	0.14	0.09	5.75	8.63	.79	136100
52	97500–97901D	5.0000	9.0000	3.3125	4.5625	0.14	0.09	5.94	8.38	.79	102600

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REQUIREMENTS:

1. Material: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with ASTM-A295, ASTM-A534, or ASTM-A535. Cage: carbon steel (one piece stamped). Powdered process steel may be used.

2. Hardness: Cones, cups and rollers: 58 to 64HRC.

3. Tolerances: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (d)			
Size range B		Tolerance	
Over	Incl.	Plus	Minus
0 3.0000	3.0000 8.0000	5 10	0 0

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000 12.0000	12.0000 24.0000	10 20	0 0

Bearing width (T)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0 4.0000	4.0000 12.0000	160 280	0 200

Assembled bearing maximum radial runout		
Cup O.D. (D)		Tolerance
Over	Incl.	
0	24.0	20

4. Operating temperature: Recommended operating temperature not to exceed 230°F.

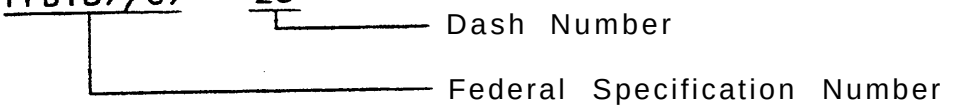
5. Lubrication: Bearings shall be furnished without lubrication. All bearings shall be preserved in accordance with MIL-P-197.

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6. Contact angle: All bearings are steep angle bearings having a contact angle (a) between 22 and 31 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

7. Part number: The part number consist of the letters FFB, the basic federal specification number, the specification sheet number and a dash number from table I.

EXAMPLE: FFB187/07 - 26



8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

9. Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

NOTE: Inch units can be converted to soft metric units as follows:

Inch	X	25.4	=	millimeters

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INTERCHANGEABILITY TABLE

MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS
MS19087-1	FFB187/07-02
MS19087-2	FFB187/07-06
MS19087-3	FFB187/07-12
MS19087-4	FFB187/07-17
MS19087-5	FFB187/07-27
MS19087-6	FFB187/07-32
MS19087-7	FFB187/07-38
MS19087-8	FFB187/07-42
MS19087-9	FFB187/07-46
MS19087-10	FFB187/07-51
MS19087-11	FFB187/07-14
MS19087-12	FFB187/07-22
MS19087-13	FFB187/07-24
MS19087-14	FFB187/07-35
MS19087-15	FFB187/07-36
MS19087-16	FFB187/07-37
MS19087-17	FFB187/07-40
MS19087-18	FFB187/07-43
MS19087-19	FFB187/07-44
MS19087-20	FFB187/07-45
MS19087-21	FFB187/07-47
MS19087-22	FFB187/07-49
MS19087-23	FFB187/07-50

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